

CO3_AT3_MCQ(Bayesian and Computational Learning)

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* Indicates required question

MCQ

19. In a Bayesian Belief Network, a node normally represents: *

- ☐ A random variable
- ☐ A training iteration
- ☐ A distance measure
- ☒ A program statement

24. During the Expectation (E) step, the algorithm primarily: *

- ☒ Estimates the expected values or probabilities of hidden variables
- ☐ Deletes hidden variables
- ☐ Randomly changes the dataset
- ☐ Stops the algorithm

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2. If $P(A)=0.4$, $P(B|A)=0.5$, and $P(B)=0.2$, what is $P(A|B)$? *

- ☐ 0.2
- ☐ 0.5
- ☒ 1.0
- ☐ 0.8

10. Two models have the same prediction accuracy. Model A is much more complex than Model B. According to the MDL principle, which model is generally preferred? *

- ☐ Model A
- ☒ Model B
- ☐ Both are always rejected
- ☐ Neither can be selected

6. Maximum Likelihood Estimation selects the parameter values that: *

- ☐ Minimize the training set
- ☒ Maximize the likelihood of the observed data
- ☐ Maximize model complexity
- ☐ Minimize prior probability

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20. Consider the network $\text{Rain} \rightarrow \text{WetGrass}$. What does the arrow indicate? *

- ☐ WetGrass causes Rain
- ☒ Rain directly influences WetGrass
- ☐ Rain and WetGrass are always independent
- ☐ Both variables have the same probability

12. Three hypotheses have posterior probabilities $H1=0.2$, $H2=0.6$, and $H3=0.2$. Which hypothesis has the greatest influence on the prediction? *

- ☐ H1
- ☐ H2
- ☐ H3
- ☒ All equally

25. During the Maximization (M) step, the algorithm: *

- ☒ Updates model parameters using the current estimates of hidden variables
- ☐ Removes training examples
- ☐ Selects a random hypothesis
- ☐ Calculates only classification accuracy

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9. The Minimum Description Length (MDL) principle prefers a model that: *

- ☐ Has maximum complexity
- ☒ Minimizes the combined description length of the model and data
- ☐ Always has zero training error
- ☐ Uses the maximum number of features

14. Suppose the posterior probabilities of three hypotheses are $H_1=0.1$, $H_2=0.7$, and $H_3=0.2$. Which hypothesis has the highest probability of being selected by the Gibbs algorithm? *

- ☐ H1
- ☒ H2
- ☐ H3
- ☐ All equally

3. In Bayesian learning, $P(H)$ represents: *

- ☐ Posterior probability
- ☐ Likelihood
- ☒ Prior probability
- ☐ Evidence

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16. Which of the following is a common application of the Naïve Bayes classifier? *

- ☒ Email spam classification
- ☐ Sorting numbers
- ☐ Image resizing
- ☐ File compression

1. Which of the following is the correct form of Bayes' theorem? *

- ☐ $P(A|B) = P(A)P(B)$
- ☐ $P(A|B) = P(B|A)P(A) / P(B)$
- ☒ $P(A|B) = P(A) + P(B)$
- ☐ $P(A|B) = P(A) - P(B)$

23. The two main steps of the EM algorithm are: *

- ☐ Selection and Mutation
- ☐ Encoding and Decoding
- ☒ Expectation and Maximization
- ☐ Training and Testing

5. A hypothesis is said to be consistent with a training set when it: *

- ☒ Classifies all training examples correctly
- ☐ Classifies only positive examples correctly
- ☐ Classifies only negative examples correctly

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15. The Naïve Bayes classifier assumes that features are: *

- ☒ Conditionally independent given the class
- ☐ Completely dependent
- ☐ Always numerical
- ☐ Always equally important

18. A Bayesian Belief Network is represented using a: *

- ☒ Directed Acyclic Graph
- ☐ Stack
- ☐ Queue
- ☐ Binary Search Tree

22. The EM algorithm is particularly useful when: *

- ☒ Hidden or unobserved variables are present
- ☐ No training data is available
- ☐ All variables are known
- ☐ Only binary data is available

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7. A coin is tossed 10 times and produces 7 heads. What is the maximum likelihood estimate of the probability of heads? *

- ☐ 0.3
- ☐ 0.5
- ☐ 0.7
- ☒ 1.0

4. The primary goal of concept learning is to: *

- ☒ Find a hypothesis that correctly classifies examples
- ☐ Increase the number of training examples
- ☐ Remove all negative examples
- ☐ Randomly generate hypotheses

11. The Bayes optimal classifier makes predictions by: *

- ☐ Selecting only one arbitrary hypothesis
- ☐ Considering hypotheses weighted by their posterior probabilities
- ☐ Using only prior probabilities
- ☒ Selecting the most complex hypothesis

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17. Suppose $P(\text{Spam})=0.4$ and $P(\text{Offer}|\text{Spam})=0.8$. What is the value of $P(\text{Offer}|\text{Spam})P(\text{Spam})$? *

- ☐ 0.16
- ☒ 0.20
- ☐ 0.32
- ☐ 1.20

21. EM stands for: *

- ☒ Error Minimization
- ☐ Expectation-Maximization
- ☐ Estimated Model
- ☐ Example Matching

8. A classifier observes 80 successful outcomes in 100 trials. What is the maximum likelihood estimate of the probability of success? *

- ☐ 0.08
- ☐ 0.20
- ☒ 0.80
- ☐ 1.25

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13. The Gibbs algorithm selects a hypothesis according to its: *

- ☐ Training-set size
- ☒ Posterior probability
- ☐ Number of attributes
- ☐ Classification error only

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